Divyam Ahuja

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Experience

Microsoft Noida
Software Engineer Mar 2025 - Present
Tech Stack: Python, TypeScript, Docker, Azure, React, Playwright

- Built and modernized **large-scale AI evaluation platform and infrastructure** for multimodal models and agentic workflows across the Microsoft Office platform.
- Optimized the observability platform and dashboard to analyze **1M** telemetry and evaluation records, enabling large-scale reliability, performance, and quality monitoring.
- Owned the end-to-end Responsible AI (RAI) evaluation pipeline gating model releases; optimized execution latency by **50%** through asynchronous worker pools, intelligent request batching, and custom retry algorithms.
- Hardened distributed workloads across the evaluation lifecycle against **transient service failures, authentication expiry, timeouts, and resource exhaustion**, reaching **99%** reliability and **<1%** infrastructure failure rate.
- Migrated from JSON-based configurations to typed configurations, reducing configuration errors by **25%** and improving overall ease of maintenance.
- Built and enhanced **automated failure detection, investigation and PR creation agentic workflows**, reducing manual triage effort by **95%** and improving diagnosis of infrastructure and application-level evaluation failures.
- Contributed to the internal Knowledge Base and Agentic Workflows, accelerating day-to-day platform operations and engineering velocity.

Advanced Micro Devices (AMD) Bangalore
System Software Designer Aug 2022 - Feb 2025
Tech Stack: C/C++, Java (OpenJDK), Linux perf, Python, Django, React, SQLite, Docker

- Engineered an OpenJDK performance monitoring and telemetry platform (Django, React, SQLite) to track nightly SPECjbb runs, enabling early detection and isolation of performance regressions.
- Automated systems to benchmark and tune Java nightly using machine learning models, achieving a **30%** reduction in manual effort and improved parameter accuracy.
- Applied advanced Java tuning techniques and profiled CPU/memory bottlenecks with Linux perf, yielding a **2%** throughput increase on SPECjbb benchmarks across High Performance Computing (HPC) servers.
- Refactored and containerized legacy benchmarking workflows with Docker for cross-platform execution, driving a **90%** reduction in deployment and environment errors.
- Awarded the **Spotlight Award** for exceptional contributions to automation, tuning, and nightly benchmark visualization.

Chromium Remote
Google Summer of Code Jun 2022 - Aug 2022
Tech Stack: C++, WebAssembly, WebAudio, WebWorkers

- Engineered a lock-free C++ ring buffer as a reference library for real-time audio streaming between WebAudio, WebAssembly, and WebWorkers, benchmarking multiple concurrency strategies.
- summerofcode.withgoogle.com

Skills

Programming Languages: C/C++, Python, Go (Golang), Java, Kotlin, TypeScript, Bash
Frameworks & Tools: PyTorch, Azure OpenAI, Android SDK / NDK, React/Next.js, Django/Flask, Node.js, gRPC, Emscripten, Playwright
Infrastructure & Databases: Kubernetes, Docker, Linux/POSIX, PostgreSQL, MongoDB, Redis, MySQL, SQLite, WebAssembly, WireGuard

Projects

Fastbin: A Code Sharing Webapp | HTMX, Go (Golang), gRPC, PostgreSQL, Docker, Kubernetes
- Built a high-performance code sharing backend in Go utilizing gRPC Protobuf microservices, PostgreSQL with connection pooling and optimized indexing, containerized with Docker and deployed on Kubernetes.
- git.divyam.dev/divyam/fastbin, fastbin.lab.divyam.dev

Homelab | Raspberry Pi 5, WireGuard, Kubernetes, Rancher, MetalLB
- Personal infrastructure hosting containerized services with WireGuard VPN tunneling, MetalLB load balancing, and Rancher-managed Kubernetes, enabling secure, high-availability external access.

Education

Deenbandhu Chhotu Ram University of Science and Technology Haryana, India
Bachelor of Technology (B.Tech.) Computer Science Aug 2018 - Aug 2022